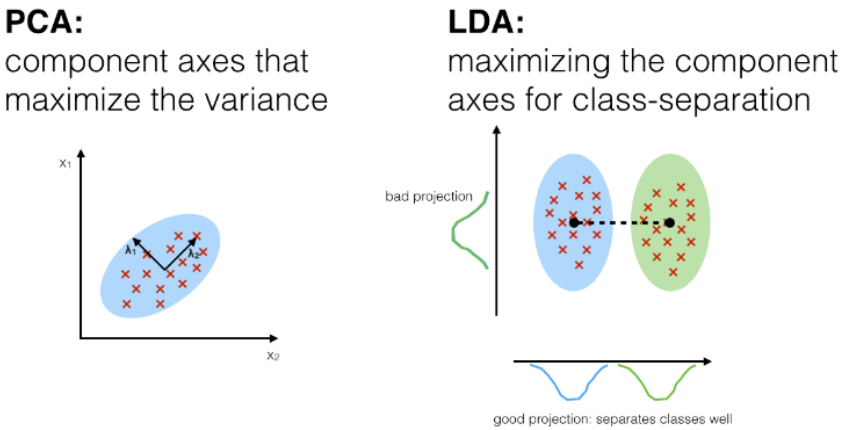
Linear Discriminant Analysis

LDA - Linear Discriminant Analysis

- **LDA (Linear Discriminant Analysis)** is a **linear model for supervised dimensionality reduction and classification technique**.
- It tries to find a new set of axes (features) that **best separate different classes**.
- **Key Idea**
  - Project high-dimensional data onto a **lower-dimensional space**
  - But in a way that **preserves class separability**

- Unlike PCA (which only cares about variance), LDA cares about **class separation**.
- LDA tries to:
  - **Maximize distance between different classes** (between-class variance)
  - **Minimize spread within the same class** (within-class variance)
- So, it finds directions where:
  - Different classes are far apart
  - Data points in the same class are tightly grouped
- Comparison between PCA and LDA:



Feature	**PCA**	**LDA**
Type	Unsupervised	Supervised
Goal	Maximize variance	Maximize class separability
Uses labels?	✗ No	☑ Yes
Components	≤ number of features	≤ classes − 1

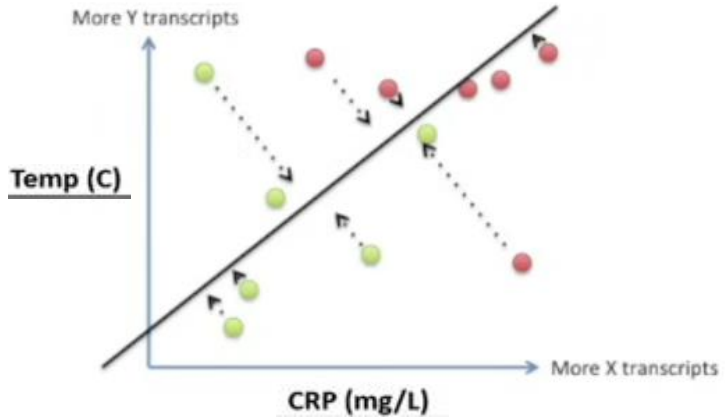
- **LDA approach in 6 steps**

1. Computes mean vectors of each class of dependent variable
2. Computes with-in class (W) and between-class (B)scatter matrices
3. Compute the Separation (S) matrix using the with-in class and between-class scatter matrices
4. Computes eigenvalues and eigenvector for S
5. Sorts the eigenvalues in descending order and select the top k
6. Obtains the new features (i.e. linear discriminants) by taking the dot product of the data and the eigenvector .

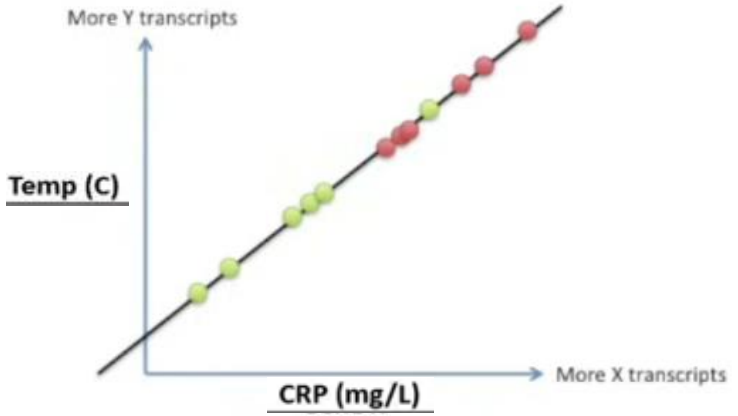
Problem: How to classify viral/ bacterial infection based on CRP and Tmp?

Infection	CRP (mg/L)	Temp (C)
Viral	40.0	36.0
Viral	11.1	37.2
Viral	30.0	36.5
Viral	21.4	39.4
Viral	10.7	39.6
Viral	3.4	40.7
Bacterial	42.0	37.6
Bacterial	31.1	42.2
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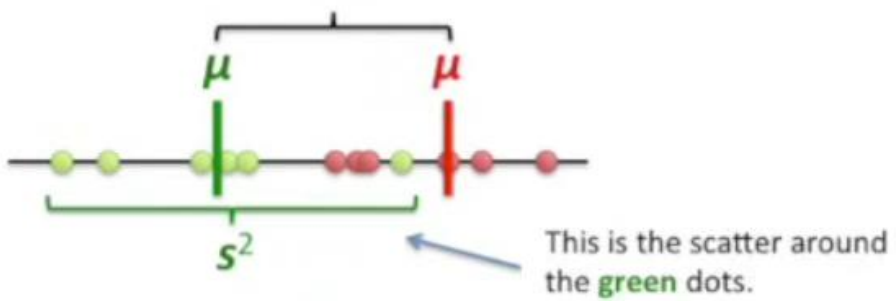


Reducing a 2-D graph to a 1-D graph with LDA



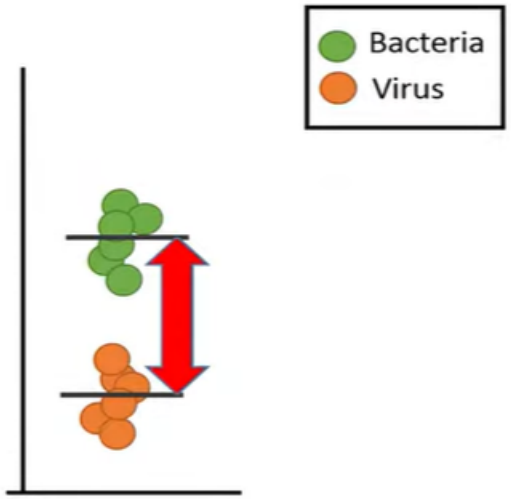
The new axis is created according to two criteria (considered simultaneously):

1) Maximize the distance between means.



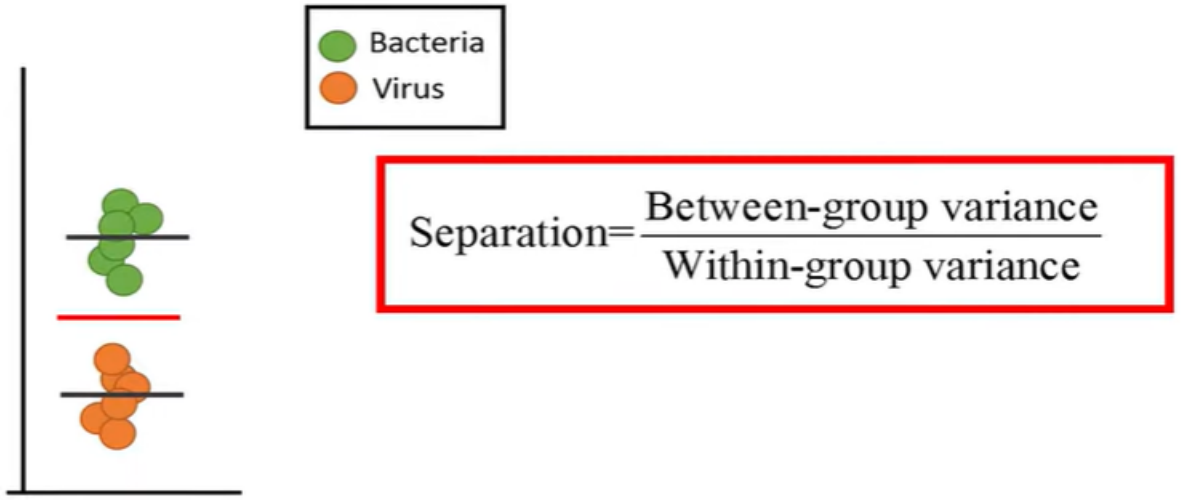
2) Minimize the variation (which LDA calls "scatter" and is represented by s^2) within each category.

Separation between the groups



s is due to that the means of the groups are far away from each other.

Separation between the groups



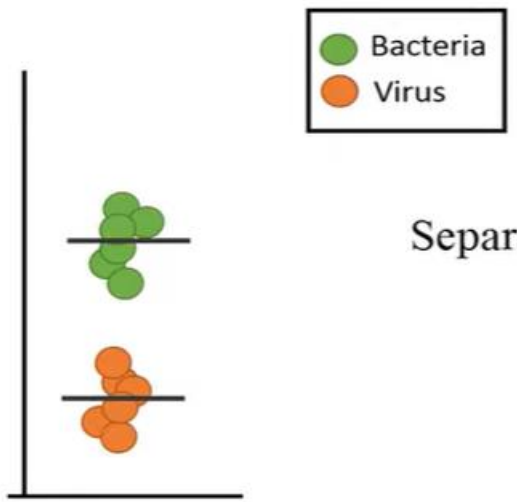
$$\text{Separation} = \frac{\text{Between-group variance}}{\text{Within-group variance}}$$

We can conclude that the separation between the groups depends on the variance between the groups, which is here called between-group variance, and the variance within the groups, the within-group variance.

The between-class scatter (variance) matrix measures the separation between the class means and the overall mean.

The within-class scatter (variance) matrix measures the variation of data points around their respective class means.

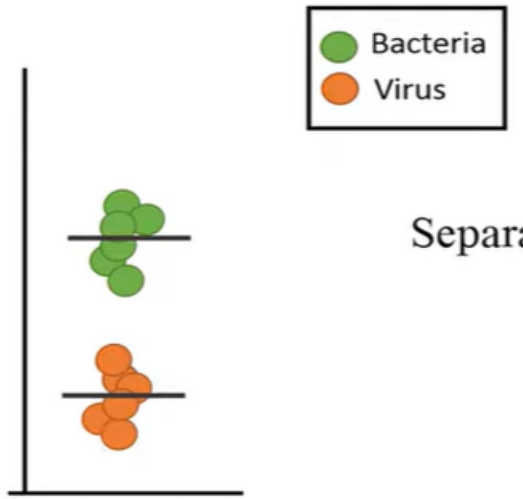
Separation between the groups



$$\text{Separation} = \frac{\text{Between-group variance}}{\text{Within-group variance}}$$

LDA combines variables based on this ratio, where the aim is to transform the data so that the between-group variance is increased and the within-group variance is reduced.

The math behind LDA



$$\text{Separation} = \frac{\text{Between-group variance}}{\text{Within-group variance}}$$

$$\boxed{S} = W^{-1}B$$

Matrix S is our target matrix. We will compute the eigenvectors of this matrix in order to get our weights.

Pooled within-group covariance matrix

$$S = W^{-1}B$$

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Viral	40.0	36.0
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Bacterial	60.4	39.4
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$$\text{cov}_{\text{virus}} = \begin{bmatrix} 188 & -21 \\ -21 & 4 \end{bmatrix}$$

$$\text{cov}_{\text{Bacteria}} = \begin{bmatrix} 228 & -24 \\ -24 & 4 \end{bmatrix}$$

$$W = \frac{\begin{bmatrix} 188 & -21 \\ -21 & 4 \end{bmatrix} + \begin{bmatrix} 228 & -24 \\ -24 & 4 \end{bmatrix}}{2} = \begin{bmatrix} 208.1 & -22.5 \\ -22.5 & 4.1 \end{bmatrix}$$

$$W = \frac{(n_1 - 1)\text{cov}(A) + (n_2 - 1)\text{cov}(B)}{n_1 + n_2 - 2}$$

If the two groups have an unequal sample size, we should use the following equation to compute the weighted average of the two covariance matrices.

Between-group covariance matrix

$$S = W^{-1}B$$

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$$T = B + W$$

$$B = T - W$$

$$T = \begin{bmatrix} 317.1 & -11.0 \\ -11.0 & 4.4 \end{bmatrix}$$

$$W = \begin{bmatrix} 208.1 & -22.5 \\ -22.5 & 4.1 \end{bmatrix}$$

This is the total covariance matrix, which shows the variance and covariance of the two variables when we do not separate the groups.

Between-group covariance matrix

$$S = W^{-1}B$$

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$$T = B + W$$

$$B = T - W$$

$$W = \begin{bmatrix} 208.1 & -22.5 \\ -22.5 & 4.1 \end{bmatrix}$$

$$T = \begin{bmatrix} 317.1 & -11.0 \\ -11.0 & 4.4 \end{bmatrix}$$

$$B = \begin{bmatrix} 317.1 & -11.0 \\ -11.0 & 4.4 \end{bmatrix} - \begin{bmatrix} 208.1 & -22.5 \\ -22.5 & 4.1 \end{bmatrix} = \begin{bmatrix} 108.9 & 11.5 \\ 11.5 & 0.32 \end{bmatrix}$$

$$S = W^{-1}B$$

$$S = \begin{bmatrix} 0.012 & 0.066 \\ 0.066 & 0.609 \end{bmatrix} \cdot \begin{bmatrix} 108.9 & 11.5 \\ 11.5 & 0.32 \end{bmatrix} = \begin{bmatrix} 2.05 & 0.16 \\ 14.15 & 0.96 \end{bmatrix}$$

$$W = \begin{bmatrix} 208.1 & -22.5 \\ -22.5 & 4.1 \end{bmatrix}$$

$$W^{-1} = \begin{bmatrix} 0.012 & 0.066 \\ 0.066 & 0.609 \end{bmatrix}$$

$$B = \begin{bmatrix} 108.9 & 11.5 \\ 11.5 & 0.32 \end{bmatrix}$$

The math behind LDA

$$S = W^{-1}B$$

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$$S = \begin{bmatrix} 2.05 & 0.16 \\ 14.15 & 0.96 \end{bmatrix}$$

Once we have calculated matrix S, the rest of the calculations are similar to those used in PCA. However, instead of calculating the eigenvalues and eigenvectors of the covariance matrix, we will calculate these on the following matrix.

The math behind LDA

$$S = W^{-1}B$$

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$$S = \begin{bmatrix} 2.05 & 0.16 \\ 14.15 & 0.96 \end{bmatrix}$$

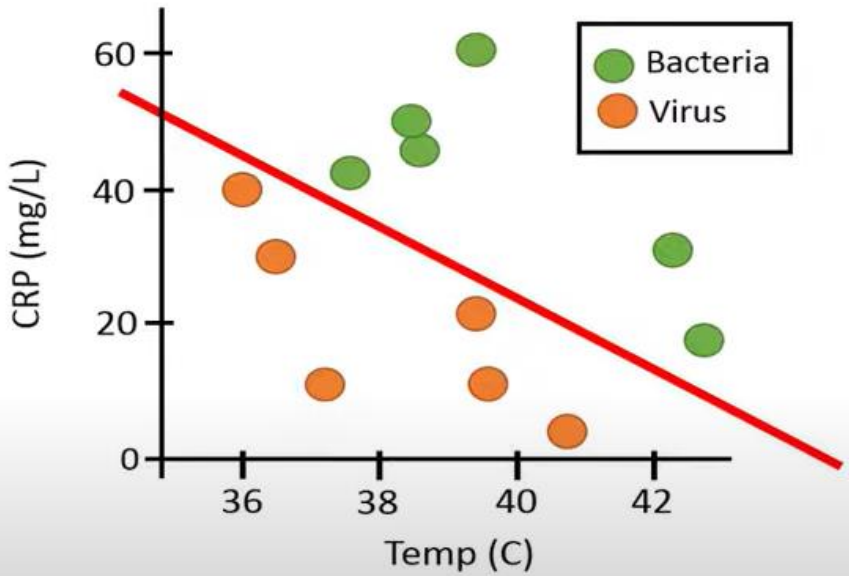
$$\text{Eigenvectors} = \begin{bmatrix} 0.150 & -0.074 \\ 0.989 & 0.997 \end{bmatrix}$$

$$\text{LD1} = 0.150 \cdot \text{CRP} + 0.989 \cdot \text{Temp}$$

By using this equation, with the given weights, we can calculate the discriminant scores that will show the best linear separation between the groups.

Separation

Infection	CRP (mg/L)	Temp (C)
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we can see that the following line can separate the two groups completely. By combining the two variables, we can get a better separation between the two groups compared to if we use one of the variables alone.

- **Why LDA?**
  - Logistic regression is a classification algorithm traditionally used for linear classification, but generally used for two-class classification problems.
  - If you have more than two classes then Linear Discriminant Analysis is the preferred linear classification technique.
  - LDA can also be used in data preprocessing to reduce the number of features just as PCA

Linear discriminant analysis (LDA) - simply explained - YouTube

StatQuest: Linear Discriminant Analysis (LDA) clearly explained. – YouTube

Linear Discriminant Analysis (sebastianraschka.com)

Inverse of a 2x2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{Inverse exists only if } ad - bc \neq 0$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Note:
For any n×n matrix: **Gauss–Jordan elimination**